Fe-C Problem 3 | Problems T4_PhaseDiagrams II.pdf #3 | Fe-C · hypereutectoid steel

Phases and Microconstituents of a 1.5 % C Steel

Original statement · Problems T4_PhaseDiagrams II.pdf, p. 2

3.- For an alloy Fe-1.5%C, at the following temperatures: T_1 = (1147+ΔT)ºC; T_2 = (1147-ΔT)ºC; T_3 = 900ºC; T_4 = (727+ΔT)ºC; T_5 = (727-ΔT)ºC. a) Indicate the phases present, their chemical composition and their mass fraction. b) Indicate the micro constituents, their chemical composition and their mass fraction at T_5.

Official Solution: a) T_1 **and** T_2**:** C_γ **= 1.5% C,** w_γ **= 1;** T_3**:** C_{Fe_3C} **= 6.7% C,** C_γ **= 1.3%** **C,** w_γ **= 0.96,** w_{Fe_3C} **= 0.04;** T_4**:** C_γ **= 0.76% C,** C_{Fe_3C} **= 6.7% C,** w_γ **= 0.87,** w_{Fe_3C} **= 0.13;** T_5**:** C_α **= 0.022% C,** C_{Fe_3C} **= 6.7% C,** w_α **= 0.78,** w_{Fe_3C} **= 0.22; b) Pearlite** $w_{Fe_3C'}$ **= 0.13,** $w_{pearlite}$ **= 0.87,** $w_{\alpha E}$ **= 0.78,** $w_{Fe_3C,E}$ **= 0.09**

PHASE 1 · HYPOTHESES & PARAMETERS

1. Equilibrium cooling, Fe-Fe3C diagram [Session 9 Slide 4].
2. $0.76 < C_0 = 1.5 < 2.14$: a **hypereutectoid steel**; at 1147 ± ΔT it is single-phase austenite (1.5 < 2.14).
3. **Reading of the figure:** at 900 °C the A_{cm} line is at about 1.3 % C (the key's value; a straight line from (0.76, 727) to (2.14, 1147) would give 1.33).

PHASE 2 · FRAMES & STRATEGY

Frame: the Fe-Fe3C diagram (wt% C, °C); one tie line per temperature. Below A_{cm} proeutectoid cementite precipitates from the austenite [Session 9 Slide 19]; at 727 °C the remaining austenite (0.76 % C) becomes pearlite.

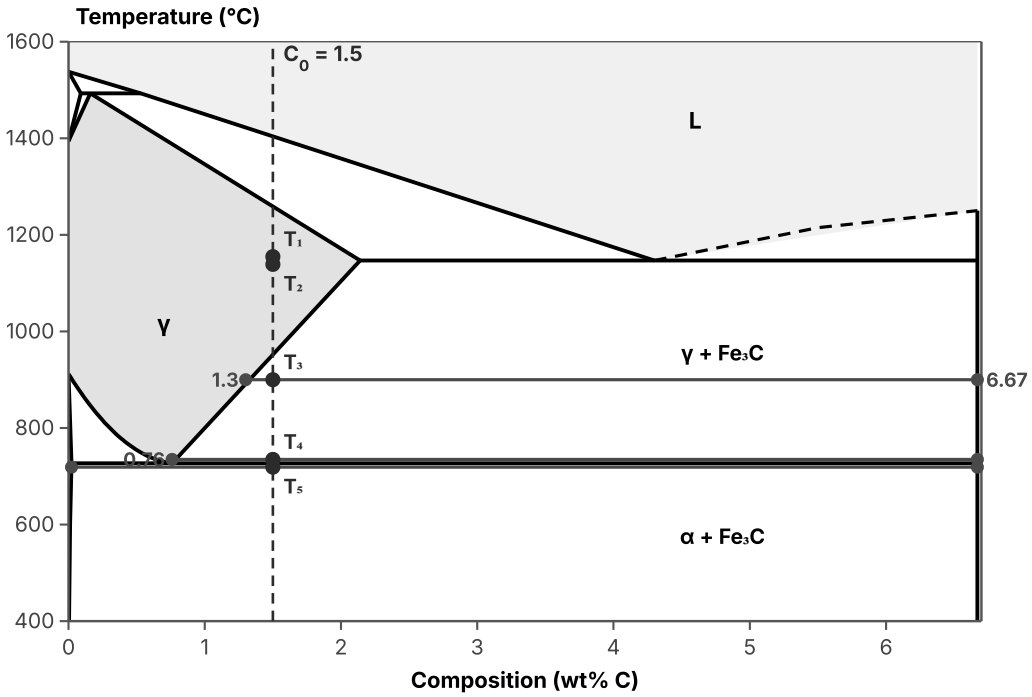


Figure 31. Fe-1.5 % C: γ at 1147 ± ΔT; γ + Fe_3C at 900 °C (γ of about 1.3 % C, a reading) and just above 727 °C; α + Fe_3C below 727 °C.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases T_1, T_2: γ, 1.5 % C, $W_\gamma = 1$.

T_3 = **900 °C** (γ 1.3 / Fe_3C 6.67; length 5.37):

$$W_\gamma = \frac{6.67 - 1.5}{6.67 - 1.3} = \frac{5.17}{5.37} = 0.963 \qquad W_{Fe_3C} = \frac{1.5 - 1.3}{5.37} = \frac{0.2}{5.37} = 0.037$$

T_4 (γ 0.76 / Fe_3C 6.67; length 5.91):

$$W_\gamma = \frac{6.67 - 1.5}{5.91} = \frac{5.17}{5.91} = 0.875 \qquad W_{Fe_3C} = \frac{1.5 - 0.76}{5.91} = \frac{0.74}{5.91} = 0.125$$

T_5 (α 0.022 / Fe_3C 6.67; length 6.648):

$$W_\alpha = \frac{6.67 - 1.5}{6.648} = \frac{5.17}{6.648} = 0.778 \qquad W_{\mathrm{Fe_3C}} = \frac{1.5 - 0.022}{6.648} = \frac{1.478}{6.648} = 0.222$$

b) Microconstituents at T_5 Proeutectoid cementite = the cementite at T_4, $W_{\mathrm{Fe_3C'}} = 0.125$; pearlite = the austenite at T_4, $W_{\mathrm{pearlite}} = 0.875$. Inside the pearlite: $W_{\alpha,E} = 0.778$ (all the ferrite) and $W_{\mathrm{Fe_3C},E} = 0.222 - 0.125 = 0.097$.

PHASE 4 · CHECKS & INTERPRETATION

- $0.778 + 0.097 = 0.875 = W_{\mathrm{pearlite}}$; and $0.875 \times 0.889 = 0.778$.
- Mass balance at T_5: $0.778 \times 0.022 + 0.222 \times 6.67 = 0.017 + 1.481 = 1.50$ % C .
- The proeutectoid cementite grows from 0.037 at 900 °C to 0.125 at 727 °C as the austenite loses carbon along A_{cm}.
- **Limiting cases** (fractions dimensionless, compositions in wt% C): at $C_0 = 0.76$ the proeutectoid Fe3C would be 0 and at $C_0 = 6.67$ the alloy would be all Fe3C; 1.5 % C lies near the eutectoid end, hence only 0.125 of proeutectoid cementite.

Check against the official solution

Agrees with the key, except for the last figure.

DISCREPANCY WITH THE OFFICIAL SOLUTION

The key gives $w_{Fe_3C,E} = 0.09$. Exactly, $0.222 - 0.125 = 0.097$ (≈ 0.10), confirmed by $0.875 \times 0.111 = 0.097$; the key subtracts the rounded values, $0.22 - 0.13 = 0.09$. The computed value is kept.